ECO 310 Midterm 1 — Preparation Strategy

March 5, 2026 · In-class · 1 page cheat sheet (front & back)

Topic Frequency Table (2022–2025 Practice Exams)

Each checkmark means the topic appeared as a graded question (MC or short answer) on that exam. Priority is based on frequency across all four exams.

Topic / Concept	'22	'23	'24	'25	Freq	Priority
Marshallian demand derivation	✓	✓	✓	✓	4/4	MUST-KNOW
EU expression setup	✓	✓	✓	✓	4/4	MUST-KNOW
EU optimization via FOC	✓	✓	✓	✓	4/4	MUST-KNOW
Risk aversion (definition/test)	✓	✓	✓	✓	4/4	MUST-KNOW
Cobb-Douglas magic rule	✓	✓	✓	✓	4/4	MUST-KNOW
Perfect complements demand	✓	✓	✓	✓	4/4	MUST-KNOW
WARP / Revealed preference	✓	✓	✓	✓	4/4	HIGH
Sub/Income effect decomposition		✓	✓	✓	3/4	HIGH
Certainty equiv. / Risk premium		✓	✓	✓	3/4	HIGH
Corner solutions / non-negativity	✓	✓	✓	✓	4/4	HIGH
Convexity of preferences (MRS)	✓	✓	✓		3/4	HIGH
Expenditure fn / Hicksian demand			✓	✓	2/4	HIGH
Portfolio allocation (risky+safe)				✓	1/4	HIGH
Rationality (complete + transitive)	✓	✓	✓		3/4	HIGH
Perfect substitutes demand	✓	✓		✓	3/4	MEDIUM
CES utility / limits				✓	1/4	MEDIUM
Arrow-Pratt measures (ARA/RRA)				✓	1/4	MEDIUM
Stochastic dominance (FOSD/SOSD)					0/4	MEDIUM
Edgeworth box / Gen. equilibrium					0/4	MEDIUM
Quasilinear utility	✓				1/4	MEDIUM

Repeated questions across exams:

- Apple tax + \$5 subsidy (Slutsky compensation) — appeared in **both 2023 and 2024**
- Framing paradox (consequentialism) — appeared in **both 2023 and 2024**
- 50-50 \$0/\$200 gamble preference — appeared in **both 2023 and 2024**

- Perfect complements sub/income effect — 2023 (sub effect = 0) and 2024 (income effect = −3)

Study Plan: Cheat Sheet + Practice Exams

Phase 1: Cheat Sheet Familiarity

1. **Read through the entire cheat sheet** end to end. For each yellow-highlighted formula, write it from memory on scratch paper, then check.
2. **Build a mental map:** Know which section contains which formula. The exam is timed—you cannot afford to search.
3. **Drill the recipes:** Section 9 of the cheat sheet has step-by-step recipes for demand derivation and EU problems. Practice following each recipe on a blank page until the steps are automatic.

Phase 2: Practice Exam Drilling

1. **Do 2025 exam first** (most recent, likely closest to your exam's style). Time yourself—treat it as real. Use only your cheat sheet.
2. **Then do 2024, 2023, 2022** in reverse chronological order.
3. After each exam, **grade yourself honestly**. For every mistake, identify:
   - Was it a **knowledge gap**? (Formula/concept not on cheat sheet or not understood)
   - Was it a **procedure error**? (Forgot to check corners, wrong inequality direction, etc.)
   - Was it a **reading error**? (Misidentified states, confused "spend" with "buy", etc.)
4. **Update your cheat sheet** if you discover a gap.

Phase 3: Targeted Weakness Drilling

1. Collect every problem you got wrong or were slow on. Re-do them from scratch.
2. For each MUST-KNOW topic, you should be able to set up and solve a problem in under 3 minutes for MC, under 10 minutes for short answer.
3. Practice the EU optimization problems heavily—they appear on every exam and are the most algebra-intensive.

Problem Patterns to Drill

Pattern	What You Must Be Able To Do	Practice Sources
Cobb-Douglas demand	Apply magic rule instantly. Normalize exponents. State spending shares.	2023 MC2, 2024 MC3, 2025 MC1
Perfect complements	Set min args equal. Plug into budget. Know sub effect = 0.	2023 MC4, 2024 MC5, 2025 MC3
Non-standard utility demand	Compute MRS from scratch. Set MRS = price ratio + budget. Check corners.	2022 Q3, 2023 Q3
WARP check	Write cost comparisons at both price vectors. Determine violation range.	2024 MC9, 2025 MC2, HW1 Prob 1
EU setup + FOC	ID states/probs/wealth. Write EU. Differentiate. Solve. Check corners.	2022 Q4, 2023 Q4, 2024 Q2b, 2025 SA2–SA3
CE / Risk premium	Compute EU, invert u to get CE, then $RP = E[W] - CE$.	2024 Q2a, 2025 SA2a
Sub/Income decomp	Find original, compensated, and new bundles. Sub = comp − orig. Inc = new − comp.	2023 MC4, 2024 MC5, 2025 MC6
Hicksian demand / EMP	Minimize expenditure s.t. utility $\geq \bar{u}$. Not the same as Marshallian.	2024 Q3d, 2025 MC3
Proof-style questions	Risk aversion comparisons, DARA implications, EU inequality proofs.	2023 Q2a, 2025 SA2a-ii, SA2b

Common Exam Traps and Mistakes to Avoid

	Trap	How to Avoid It
1	**Marshallian $\neq$ Hicksian**	Read the question. "Cheapest bundle to reach utility $\bar{u}$" = EMP/Hicksian. "Best bundle given income m" = UMP/Marshallian. The 2024 solutions note this cost many students half credit.
2	**Perfect comp: sub effect = 0**	With min{}, the compensated bundle equals the original bundle. The entire price-change effect is income effect. Do not compute a "substitution effect."
3	**WARP inequality directions**	"A revealed preferred to B" means A was chosen *and* B was affordable (cost of $B \leq$ cost of A at those prices). Draw a table of costs.
4	**EU: wrong wealth per state**	Use a probability tree. Final wealth = initial wealth $\pm$ all gains/losses in that state. Write it out explicitly before computing EU.
5	**Forgetting non-negativity**	After solving the FOC, always check $x_i \geq 0$. If negative, set that good to 0 and re-solve. Especially common with quasilinear and perfect substitutes.
6	**"Spend $k\times$" $\neq$ "Buy $k\times$"**	Cobb-Douglas magic rule gives *spending* shares. "Spends 3 times as much on good 2" $\neq$ "buys 3 times as many units" unless prices are equal. Tested explicitly in 2024 MC3.
7	**CD exponents not normalized**	$u = x^a y^b$: spending share on x is $\frac{a}{a+b}$, NOT just a. Always divide by sum of exponents.
8	**vNM: only affine transforms**	Expected utility rankings are preserved only by $\tilde{u} = a + bu$ ($b > 0$). General monotone transforms change EU rankings. Ordinal utility allows any monotone transform; cardinal (vNM) does not.
9	**Variance $\neq$ Std deviation**	Portfolio std dev = $\alpha \cdot \sigma_{\text{risky}}$. If problem gives variance, take the square root first. Tested in 2025 MC4.
10	**Quasilinear: income effect for x_1**	At interior, demand for x_1 is independent of m. But if m is too low, you hit a corner at $x_2 = 0$, and then $x_1 = m/p_1$ *does* depend on income.
11	**Risk aversion $\neq$ always choose certain**	A risk-averse person prefers \$100 certain over 50-50 \$0/\$200 (same EV). But \$90 certain vs. 50-50 \$0/\$200 is **ambiguous**—lower EV but no risk. Cannot determine without knowing the utility function.
12	**CES limit behavior**	As $r \to 0$: Cobb-Douglas. As $r \to -\infty$: perfect complements. As $r = 1$: perfect substitutes. Know which limit is which.

Bottom line: Every exam has (1) a Cobb-Douglas/complements demand MC, (2) a WARP question, (3) a non-standard demand derivation, and (4) an EU optimization problem. Master these four patterns and you cover >80% of the exam.